

I. Periodization and Discretization

We know that ① $\sum_{n=-\infty}^{+\infty} \delta(t - nT_0) \xleftrightarrow{F} \sum_{k=-\infty}^{+\infty} \frac{2\pi}{T_0} \delta(\omega - k\omega_0)$

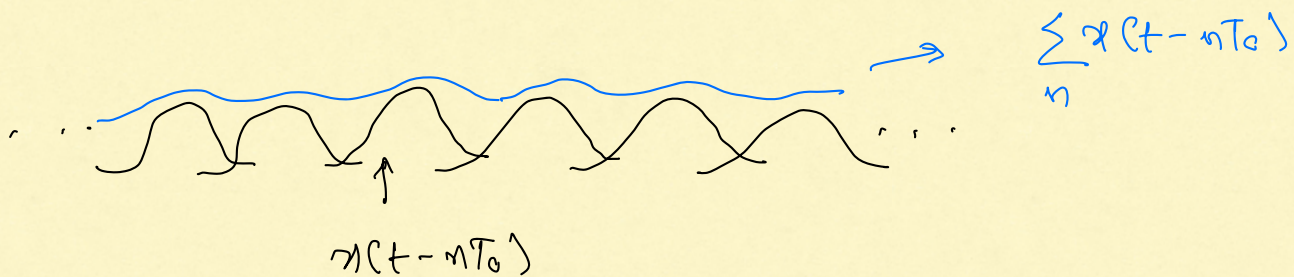
where $\omega_0 = \frac{2\pi}{T_0}$.

② $x(t) \cdot T(t) \xleftrightarrow{F} \frac{1}{2\pi} X(\omega) * \hat{T}(\omega)$

③ $x(t) * T(t) \xleftrightarrow{F} X(\omega) \cdot \hat{T}(\omega)$.

Periodization:

$x(t) \rightarrow$ any well-behaved signal, non-periodic.



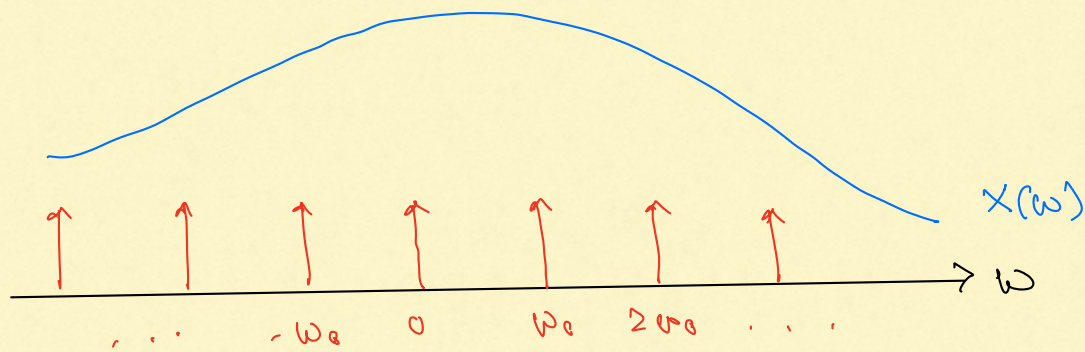
$$y(t) = x(t) * \sum_n \delta(t - nT_0) = \sum_n x(t) * \delta(t - nT_0)$$

$$y(t) = \sum_n x(t - nT_0)$$

$\therefore$ Convolution with Dirac comb will periodize a signal.

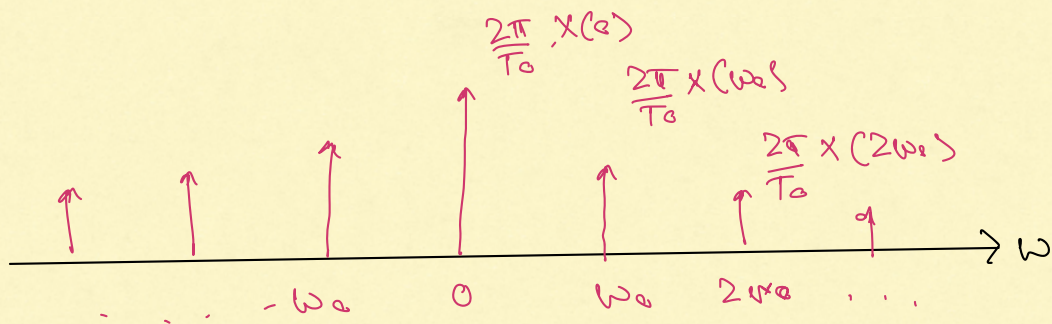
$$Y(\omega) = X(\omega) \cdot \sum_k \frac{2\pi}{T_0} \delta(\omega - k\omega_0)$$

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$$= \sum_k \frac{2\pi}{T_0} X(\omega) \cdot \delta(\omega - k\omega_0)$$

$$= \sum_k \frac{2\pi}{T_0} X(k\omega_0) \delta(\omega - k\omega_0)$$



Periodization in time domain $\Rightarrow$ Discretization in Freq. Domain.

Connection to Fourier Series:

Since $y(t)$ is periodic with period T_0 , y has a

Fourier series:

$$y \overset{\text{F.S.}}{\longleftrightarrow} c_k.$$

Assuming convergence of Fourier series:

$$y(t) = \sum_{-\infty}^{+\infty} c_k \cdot e^{jk\omega_0 t}, \quad \omega_0 = \frac{2\pi}{T_0}$$

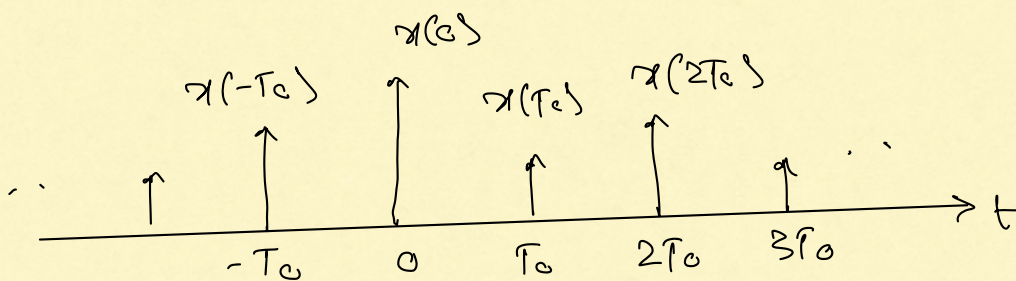
$$\begin{aligned} Y(\omega) &= \sum_k F(c_k \cdot e^{jk\omega_0 t}) \\ &= \sum_k c_k \cdot \delta(\omega - k\omega_0) \cdot 2\pi \end{aligned}$$

Discretization (in time domain) / Sampling:

Suppose we want to sample a signal $x(t)$ at integer multiples of T_0 , $T_0 > 0$. We call T_0 : sampling interval or sampling period.

The sample values: at time nT_0 , value $x(nT_0)$.
This information is represented as:

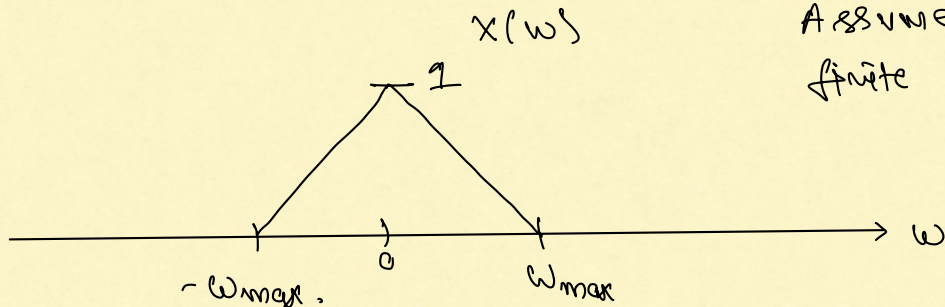
$$y(t) = \sum_n x(nT_0) \cdot \delta(t - nT_0)$$



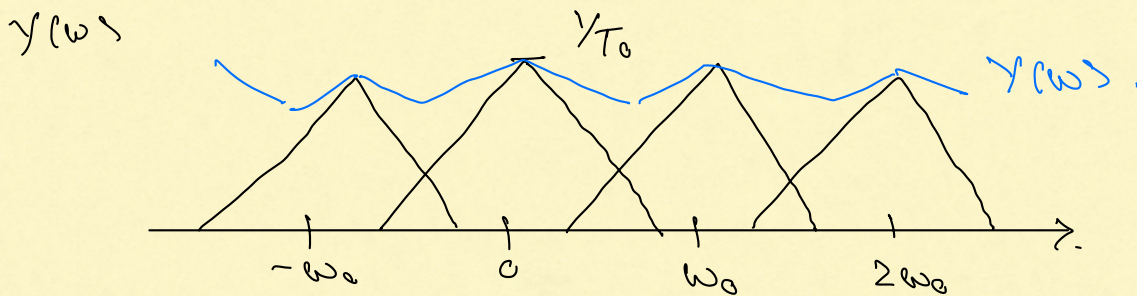
$$\begin{aligned} y(t) &= x(t) \cdot \sum_n \delta(t - nT_0) \\ &= \sum_n x(t) \cdot \delta(t - nT_0) \\ &= \sum_n x(nT_0) \cdot \delta(t - nT_0) \end{aligned}$$

$$Y(\omega) = \frac{1}{2\pi} \left[X(\omega) * \frac{2\pi}{T_0} \sum_k \delta(\omega - k\omega_0) \right], \quad \omega_0 = \frac{2\pi}{T_0}$$

$$= \frac{1}{T_0} \sum_k X(\omega - k\omega_0).$$



Assume X has a finite bandwidth.

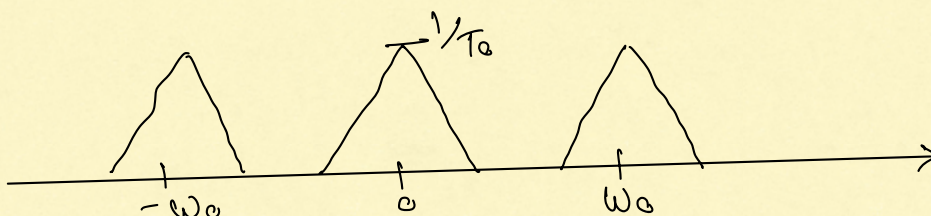


Hence, subsequent copies of $X(\omega - k\omega_0)$ overlap with each other, and hence, this causes a phenomenon called

aliasing:

That, with no further information available about $x(t)$, apart from its bandwidth, we cannot recover $x(t)$ from the sampled signal/distribution $y(t)$.

On the contrary, if $\omega_0 > 2\omega_{\max}$:

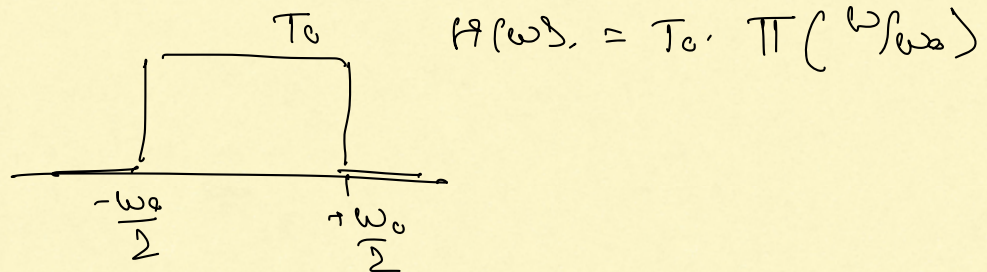
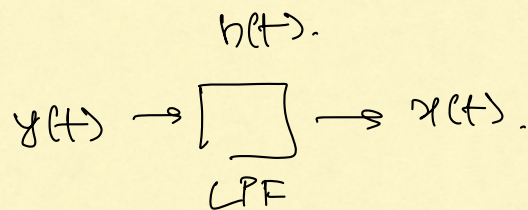


The subsequent copies of $x(\omega - k\omega_0)$ do not overlap with each other.

How do we reconstruct $x(t)$ from its sample values:

$x(nT_0)$, $n \in \mathbb{Z}$, or equivalently, from $y(t)$?

Idea:



Home Work: $\text{sinc}(t/T_0) \xleftrightarrow{F} H(\omega) = T_0 \cdot \Pi(\omega/\omega_0)$

$$\Rightarrow x(t) = y(t) * \text{sinc}(t/T_0).$$

$$= \sum_n x(nT_0) \cdot \delta(t - nT_0) * \text{sinc}(t/T_0)$$

$$= \sum_n x(nT_0) \text{sinc}\left(\frac{t - nT_0}{T_0}\right).$$